They want a brand new, modern client with features like a contacts list, and cool tricks: for instance, when you add a new recipient to an email, your application should check if it's already in the contacts list, and if it's not, a popup (like the one shown in figure 4.1) should appear asking you if you'd like to add the new recipient.

And, needless to say, you are the lucky one that gets to be responsible for implementing this feature.

Of course, the resources allocated for the project are scarce, so you are only going to do a refactoring of the client, while on the server side you'll have to make do with legacy code and legacy services running on proprietary machines.

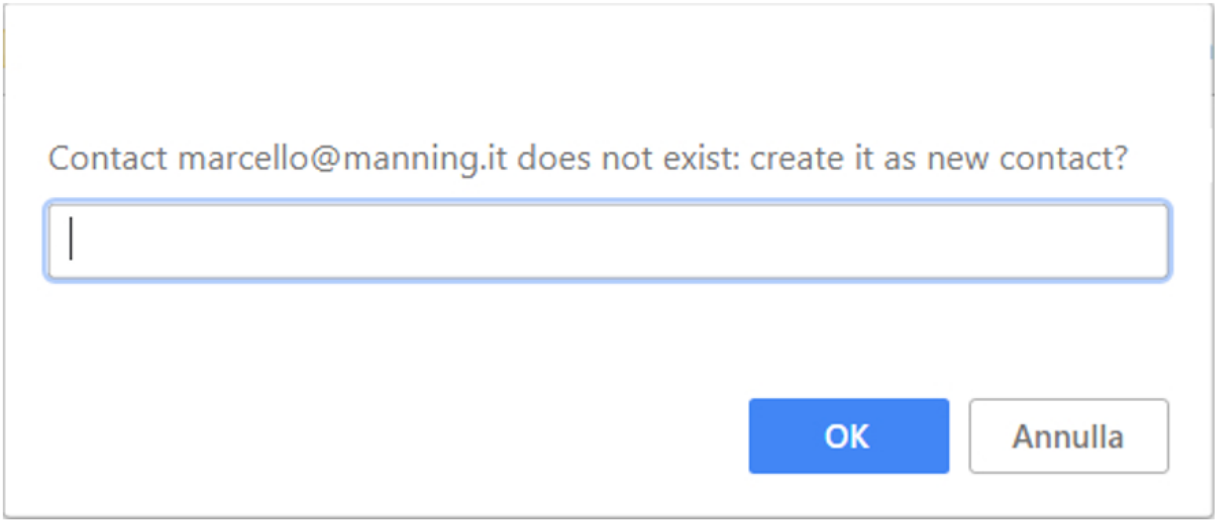


Figure 4.1 Our email application prompting the user to add a new contact after an unsuccessful search.

Calling the database every time you need to check an email address against your contacts list is out of question: you are stuck with this legacy machine that can't scale up, and you didn't get funds for refactoring and scaling it out[1]. Your DB